

NOAH M. SWAN

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EDUCATION & SKILLS

University of Chicago, Chicago, IL
Master of Science in Statistics

Expected May 2027

Villanova University, Villanova, PA

May 2023

Bachelor of Science in Mathematics, Minors in Economics, Statistics and French, Honors PPE Concentration
Graduated Summa Cum Laude, Phi Beta Kappa Honor Society

WORK EXPERIENCE

BCG X | Chicago

Summer 2026

Data Science Intern (Forward Deployed AI Scientist)

- Built two dbt pipelines on Databricks that unified client transaction data, an external Supabase competitor feed, and runtime parameters from Postgres to produce master tables for weekly pricing statistics across 1,500 stores
- Designed a competitor benchmarking framework for pricing simulator, calibrating price indices to enable targeted, elasticity-driven price recommendations

University of Chicago Data Science Institute | Chicago

January 2026-Present

Research Assistant

- Engineered NLP pipeline using ChromaDB to automate the extraction of fish stock evaluations and source reports, building a training dataset from 140,000+ document chunks using L2 and Levenshtein distances
- Designed inference method using embedding distance metrics and LLM querying to generate fish stock evaluations from reports, replacing manual workflows and enabling scalable analysis across 1,500+ fish stocks

NERA Economic Consulting | Washington DC

July 2023-May 2025

Associate Analyst

- Drove efforts to comply with governmental data requests for multiple billion-dollar mergers, coordinating data review with researchers to provide accurate submissions to DOJ/FTC that supported regulatory approval
- Leveraged arrow and data.table packages in R to compute market shares and summary statistics on datasets up to 1.5 TB, reducing computational time by 5-20x and cutting memory usage by 50-70%
- Developed S3 method for internal R package to streamline charting with ggplot2; improved reproducibility through standardized charts and reduced build time on deliverables by 35%
- Generated 50+ service area maps using ArcMap and Python (ArcPy), highlighting geographic overlaps between merging parties to inform antitrust risk assessment and anticipate DOJ arguments
- Charted receipt data and transactional sales data to identify customer overlaps and event studies, guiding analyses of product substitutability used in regulatory submissions

Orlando City SC | Remote

Fall 2022

Soccer Analytics Intern

- Engineered a ball-progression metric to evaluate team possession play from event data using R, improving how the team assessed buildup effectiveness
- Designed framework to isolate game-state bias from player and team statistics, strengthening the ability to evaluate opponents and possible transfer targets

Oklahoma City Energy FC | Remote

2021-2022

Lead Data Analyst

- Developed ETL pipeline to receive data from Wyscout API and designed Shiny dashboard suite to track team's KPIs and support coaching strategies ahead of a top 5 league finish
- Built recruitment dashboard integrating custom filters to track and compare player performance metrics across 3 leagues, improving coaching staff's ability to identify, evaluate, and compare transfer targets

SELECTED PROJECTS

Simulating Tennis Matches for Outcome Prediction (PyTorch, XGBoost, Scikit-learn, SciPy, pandas, Matplotlib)

- Developed a Bayesian forecasting method for predicting match outcomes from simulated player statistics, achieving 11% better accuracy over a naïve rank-based prediction strategy
- Estimated per-player Beta posteriors using Beta-Binomial conjugate updates with method-of-moments empirical priors, conditioned on surface and opponent rank, to generate pre-match feature distributions for inference

MLE for Truncated Normal Distribution with Gradient Descent (C++)

- Built a C++20 implementation of maximum likelihood estimation for the truncated normal distribution using gradient descent, enabling parameter recovery in settings where closed-form estimators do not exist
- Designed a modular system with JSON-configurable parameters to support flexible experimentation and reproducible results

Marginal Effect of XGBoost Hyperparameter Tuning (XGBoost, hyperopt, polars)

- Ran XGBoost tuning experiments across search-space regimes; built a modular codebase for prediction tasks
- Found that shorter tuning runs across focused search spaces achieved comparable performance with time savings of ~37%

TECHNICAL SKILLS AND INTERESTS

Programming Languages: Python, R, C++, SQL, MATLAB

Software and Tools: pandas, polars, XGBoost, PyTorch, Scikit-learn, NumPy, SciPy, Matplotlib, ChromaDB, PySpark, tidyverse, data.table, Arrow, ggplot2, shiny, Git, ArcPy/ArcMap, Claude Code

Additional Interests: Surrealist Books, Soccer, Tennis, College Football, NBA, Sports Analytics, Art History, French